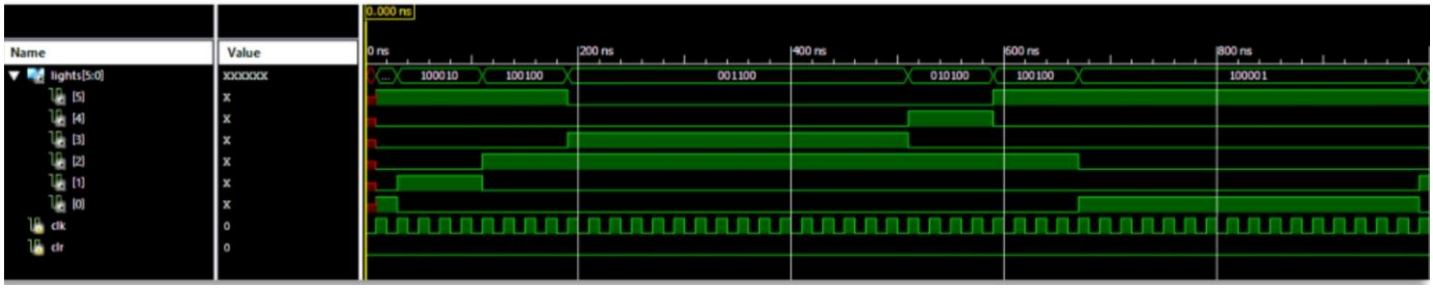
```
001  `timescale 1ns / 1ps
002
003  module Traffic_Light_Time(
004      input wire clk,
005      input wire clr,
006      output reg [5:0] lights
007      );
008
009      reg [2:0] state;
010      reg [3:0] count;
011
012      parameter S0 = 3'b000, S1 = 3'b001, S2 = 3'b010,
013                S3 = 3'b011, S4 = 3'b100, S5 = 3'b101;
014
015      parameter SEC5 = 4'b1111, SEC1 = 4'b0011;
016
017      always @(posedge clk or posedge clr)
018      begin
019          if (clr == 1)
020          begin
021              state <= S0;
022              count <= 0;
023          end
024          else
025          begin
026              case (state)
027                  S0: if (count < SEC5)
028                      begin
029                          state <= S0;
030                          count <= count + 1;
031                      end
032                      else
033                      begin
034                          state <= S1;
035                          count <= 0;
036                      end
037
038                  S1: if (count < SEC1)
039                      begin
040                          state <= S1;
041                          count <= count + 1;
042                      end
043                      else
044                      begin
045                          state <= S2;
046                          count <= 0;
047                      end
048
049                  S2: if (count < SEC1)
050                      begin
051                          state <= S2;
052                          count <= count + 1;
053                      end
054                      else
055                      begin
056                          state <= S3;
057                          count <= 0;
058                      end
059
060                  S3: if (count < SEC5)
061                      begin
062                          state <= S3;
063                          count <= count + 1;
064                      end
065                      else
066                      begin
067                          state <= S4;
068                          count <= 0;
069                      end
```

```
069                     end
070
071                 S4: if (count < SEC1)
072                     begin
073                         state <= S4;
074                         count <= count + 1;
075                     end
076                     else
077                     begin
078                         state <= S5;
079                         count <= 0;
080                     end
081
082                 S5: if (count < SEC1)
083                     begin
084                         state <= S5;
085                         count <= count + 1;
086                     end
087                     else
088                     begin
089                         state <= S0;
090                         count <= 0;
091                     end
092
093                 default:
094                     begin
095                         state <= S0;
096                         count <= 0;
097                     end
098             endcase
099         end
100     end
101
102     always @(*)
103     begin
104         case (state)
105             S0: lights = 6'b100001; // red green
106             S1: lights = 6'b100010; // red yellow
107             S2: lights = 6'b100100; // red red
108             S3: lights = 6'b001100; // green red
109             S4: lights = 6'b010100; // yellow red
110             S5: lights = 6'b100100; // red red
111             default: lights = 6'b100001; // red green
112         endcase
113     end
114
115 endmodule
```



```
`timescale 1ns / 1ps

module Traffic_Light_Sensor(
    input clk,
    input clr,
    input SNS1,
    input SNS2,
    output reg [1:0] TL1, TL2
    );

    reg [1:0] state, nextstate;

    parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b10, S3 = 2'b11;
    parameter green = 2'b00, yellow = 2'b01, red = 2'b10;

    always @(posedge clk, posedge clr)
    begin
        if (clr) state <= S0;
        else state <= nextstate;
    end

    always @(*)
    begin
        case (state)
            S0: if (SNS2) nextstate = S1;
                else nextstate = S0;
            S1: nextstate = S2;
            S2: if (SNS1) nextstate = S3;
                else nextstate = S2;
            S3: nextstate = S0;
            default: nextstate = S0;
        endcase
    end

    always @(*)
    begin
        if (state == S0)
        begin
            TL1 = green;
            TL2 = red;
        end
        else if (state == S1)
        begin
            TL1 = yellow;
            TL2 = red;
        end
        else if (state == S2)
        begin
            TL1 = red;
            TL2 = green;
        end
        else
        begin
            TL1 = red;
            TL2 = yellow;
        end
    end
endmodule
```

```
01    `timescale 1ns / 1ps
02
03    module Traffic_Light_Sensor_tb;
04        reg clk, clr, SNS1, SNS2;
05        wire [1:0] TL1, TL2;
06
07        Traffic_Light_Sensor uut (
08            .clk(clk), .clr(clr),
09            .SNS1(SNS1), .SNS2(SNS2),
10            .TL1(TL1), .TL2(TL2)
11        );
12
13        initial
14        begin
15            clr = 1;
16            #13
17            clr = 0;
18
19            SNS1 = 0; SNS2 = 0; #10;
20            SNS1 = 0; SNS2 = 1; #40;
21            SNS1 = 0; SNS2 = 0; #40;
22            SNS1 = 1; SNS2 = 0; #30;
23            SNS1 = 0; SNS2 = 0; #20;
24            SNS1 = 0; SNS2 = 1;
25        end
26
27        always
28        begin
29            clk = 1;
30            forever # (10/2) clk = ~clk;
31        end
32    endmodule
```

